Inverse Systems for LTI Systems Described by Difference Equations

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Key Concepts

1) The system function of the inverse system $H_I(z)$ is the inverse of the system function $H(z)$, i.e., $H_I(z) = \frac{1}{H(z)}$ The ROCs of the $H(z)$ and $H_I(z)$ must intersect.

2) The ROC for a stable and causal inverse system must include the unit circle.

    a) The magnitude of $H(z)$ must not be zero on the unit circle for $H_I(z)$ to converge.

    b) The zeros of $H(z)$ must not lie on the unit circle.

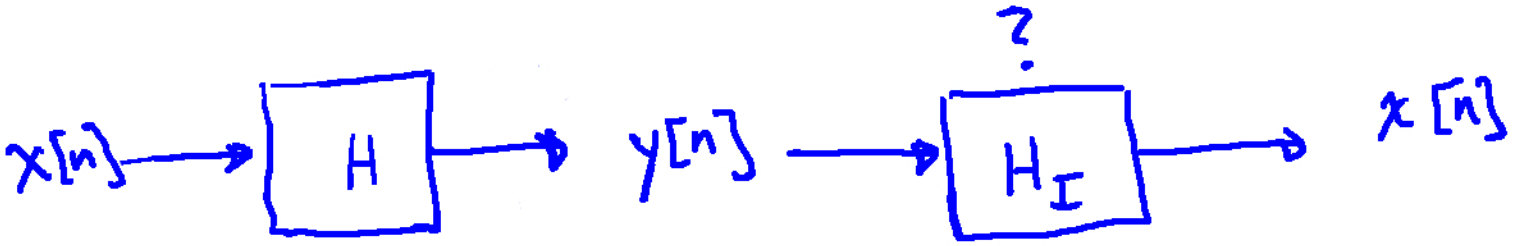
3) The zeros of $H(z)$ are the poles of $H_I(z)$.

   a) The poles of $H_I(z)$ must lie inside the unit circle for $H_I(z)$ to be both stable and causal.

   b) The zeros of $H(z)$ must lie inside the unit circle for $H_I(z)$ to be both stable and causal..

4) A stable and causal inverse $H_I(z)$ exists for a stable and causal system $H(z)$ if and only if both the poles and zeros of $H(z)$ are inside the unit circle.

Video Screenshots

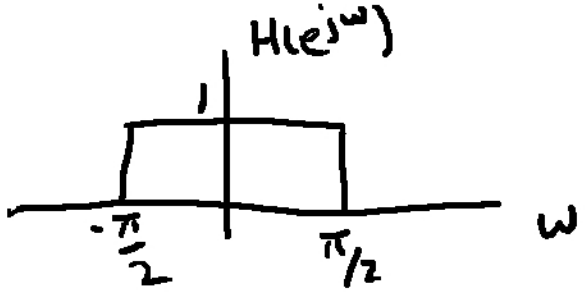


$y[n] = h[n] * x[n]$, $x[n] = h_I[n] * y[n]$ $\Rightarrow$ $h_I[n] * h[n] = \delta[n]$

$H(z)\,H_I(z) = 1 \Rightarrow H_I(z) = \frac{1}{H(z)}$ (ROCs must intersect)

For stability, ROC for $H_I(z)$ must include $|z| = 1$

$\Rightarrow |H(z)|\Big|_{z=e^{j\omega}} > 0$



$\Rightarrow$ no inverse system

LTI systems described by difference equations

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

zeros: c_k
poles: d_k

Cannot have $|c_k| = 1$ for stable $H_I(z)$

$$H_I(z) = \frac{a_0}{b_0} \frac{\prod_{k=1}^{N} (1 - d_k z^{-1})}{\prod_{k=1}^{M} (1 - c_k z^{-1})}$$

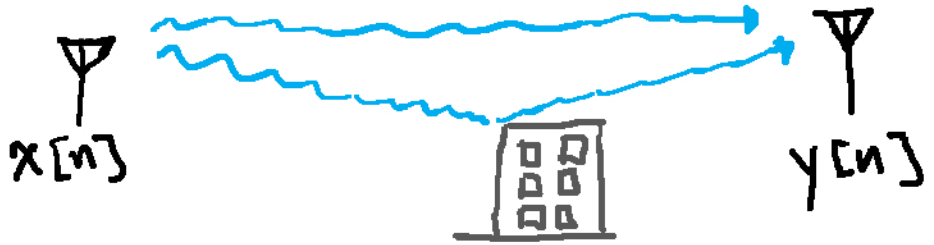
zeros: d_k
poles: c_k

ROC?

Stable/causal $H(z) \iff |d_k| < 1$

Stable/causal $H_I(z) \iff |c_k| < 1$

Example: Multipath Communication



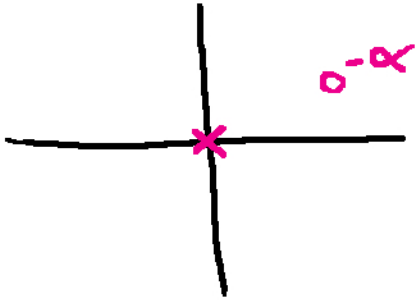
Diff. eqn model:

$$y[n] = x[n] + \alpha x[n-1]$$

Does a stable/causal inverse system exist?

$$H(z) = 1 + \alpha z^{-1}$$

pole: $z = 0$, zero: $z = -\alpha$



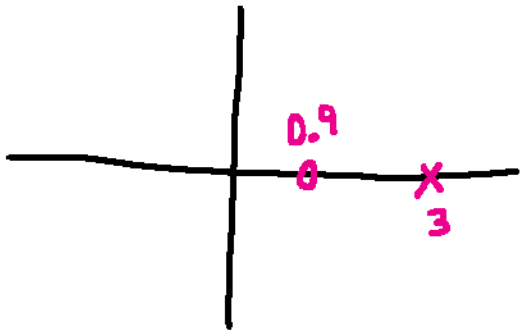
Require $|\alpha| < 1$!

$$H_I(z) = \frac{1}{1+\alpha z^{-1}} \Rightarrow y[n] + \alpha y[n-1] = x[n]$$

Example: $H(z) = \dfrac{z^{-1} - 1/3}{1 - 0.9\,z^{-1}}$ (stable/causal)

$$H_I(z) = \frac{1 - 0.9z^{-1}}{z^{-1} - 1/3} = -3\,\frac{1 - 0.9\,z^{-1}}{1 - 3z^{-1}}$$

pole at $z = 3$



ROC $|z| < 3$ stable, not causal

$|z| > 3$ causal, not stable

A stable/causal inverse system does not exist!

If $H(z) = \dfrac{z^{-1} - 3}{1 - 0.9\,z^{-1}}$, $H_I(z) = -\frac{1}{3}\,\dfrac{1 - 0.9z^{-1}}{1 - 1/3\,z^{-1}}$ pole at $z = 1/3$

stable/causal inverse system exists!

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